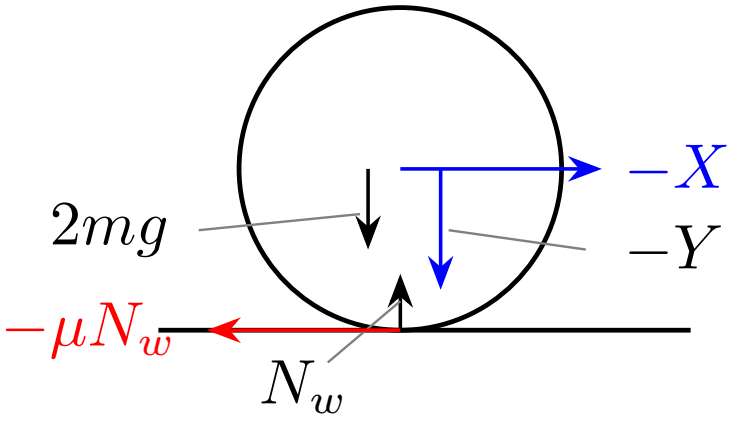
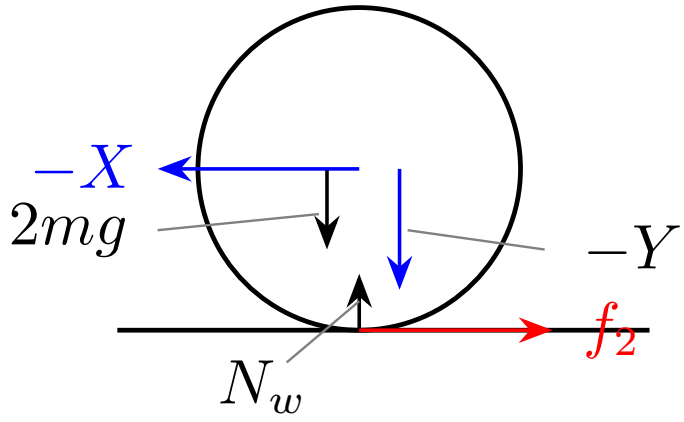
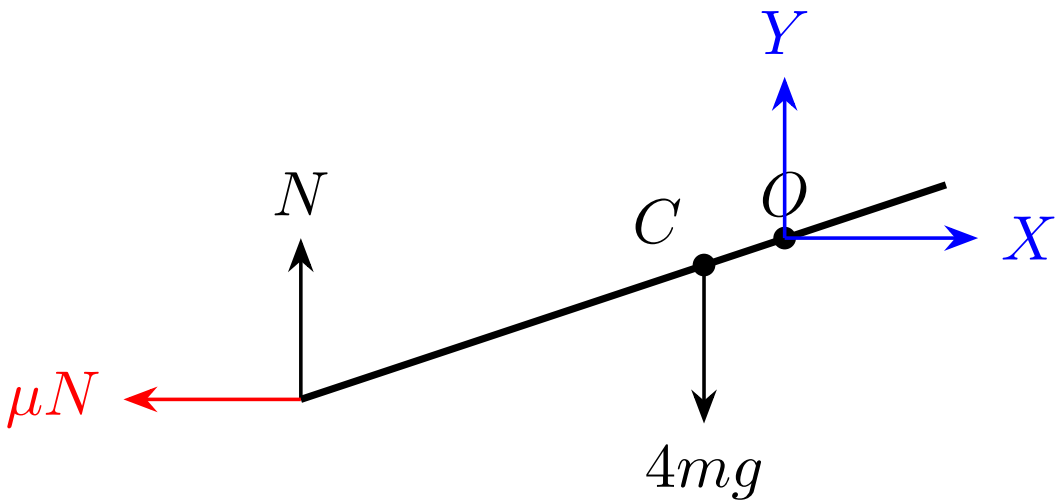
阶段 I：滑动



阶段 II：滚动





车架无角加速度，绕 C 取矩